University at Buffalo
Department of Mechanical and Aerospace Engineering
School of Engineering and Applied Sciences

Jun 3, 2026

Taro Toyoizumi, Ph.D.
Editor, *Neural Networks*

Re: Manuscript submission to *Neural Networks*

Dear Dr. Toyoizumi,

I am writing to submit a manuscript for consideration for publication in *Neural Networks*. The title of the manuscript is "**Evidence-Guided Neural Architecture Selection under Uncertainty for Subject-Specific Blood Glucose Forecasting**." This work has not been published previously and is not under consideration elsewhere.

This manuscript presents EVIDENT, a framework for discovering and validating Bayesian Temporal Convolutional Network (B-TCN) architectures under uncertainty. EVIDENT uses hierarchical Bayesian inference and model evidence to identify architectures that optimally balance predictive accuracy and complexity. Demonstrated on blood-glucose forecasting problems, the framework systematically rejects both under- and over-parameterized models, identifies architectures that generalize reliably to unseen patients, and naturally supports plausibility-weighted model averaging when multiple architectures are competitive. More broadly, the work introduces a principled approach to neural architecture selection in data-limited and high-consequence settings.

Thank you for considering our submission. Please feel free to contact me with any questions or requests for additional information.

Sincerely,

Danial Faghihi

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